



MANIPAL INSTITUTE OF TECHNOLOGY

Manipal

(A constituent unit of MAHE, Manipal)

III SEMESTER B. TECH (CSFT)

PROGRAMMING PARADIGM LABORATORY(CEF2111)

LAB MID TERM EXAMINATION-2026-27

Date:04-SEPT-2026

Student Name:	Roll No:	SET NO: 1
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Duration: 2.5 Hours

Maximum Marks: 20

Problem Statement: Manipal Academy of Higher Education is organizing TechTatva Fest with the following six events:

Event Code	Event Name
1	SAAHITI
2	IKYA
3	ROBOWARS
4	CODE MARATHON
5	Alrena
6	Quantum Leap

Develop a menu-driven C program to manage participant registrations using nested structures, an array of structures, dynamic memory allocation, arrays, and user-defined functions.

Requirements

Design suitable nested structures to store:

- Participant Registration ID
- Participant name and gender
- Number of events registered
- Details of the events selected

The number of participants must not be fixed. Dynamically allocate memory for participant records using `realloc()` and release the memory using `free()` before program termination.

Use following requirements for structures:

```
#define MAX_EVENTS_PER_PERSON 3
#define MAX_PARTICIPANTS_PER_EVENT 6
#define TOTAL_EVENTS 6
```

<pre>struct Participant { int regId; struct PersonalDetails personal; int eventCount; struct EventDetails events[MAX_EVENTS_PER_PERSON]; };</pre>	<pre>struct PersonalDetails { char name[50]; char gender; };</pre>	<pre>struct EventDetails { int code; char name[30]; };</pre>
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Registration Rules: The program must enforce the following rules:

1. Each participant must have a unique Registration ID.
2. Gender must be M, F, or O.
3. A participant can register for a maximum of 3 events.
4. If a participant attempts to register for more than 3 events, display: Maximum 3 events are allowed.
5. Each event can accommodate a maximum of 6 participants.
6. If the 7th participant attempts to register for an event, display:
Event is FULL.

Maximum capacity of 6 participants has been reached.

7. A participant cannot register for the same event more than once.
8. Event code must be between 1 and 6.
9. If one selected event is full, the participant should be allowed to select another available event.
10. Statistics must be calculated by traversing the participant array.
11. Do not maintain a separate array for event-wise registration counts.

Implement the Following Operations

1. Register Participant

Accept the participant's Registration ID, name, gender and selected events.

While registering, check:

- Duplicate Registration ID
- Valid gender
- Maximum 3 events per participant
- Valid event code
- Duplicate event selection
- Maximum 6 participants per event

Display appropriate messages for invalid attempts.

2. Display Participants

Display the details of all registered participants, including:

- Registration ID
- Name
- Gender
- Number of events
- Names of registered events

3. Gender-wise Statistics

Calculate and display:

- Number of Male participants
- Number of Female participants
- Number of Other participants
- Total participants

A participant must be counted only once, irrespective of the number of events registered.

4. Event-wise Statistics

Traverse the participant array and calculate the number of registrations for each event.

Display:

Event	Registered	Capacity	Available	Status
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The capacity of every event is 6.

If registered participants = 6, display the status as: FULL

5. Maximum Event Registrant

Find the maximum number of events registered by any participant and display **all participants** having that maximum.

6. Exit

Before termination:

- Release all dynamically allocated memory using free().
- Display an appropriate termination message.

Function Requirements: Implement at least five user-defined functions.

Your program should demonstrate the following function categories wherever appropriate:

1. No arguments, no return value
2. No arguments, return value
3. Arguments, no return value

4. Arguments, return value

Suggested functions:

```
void displayMenu();  
void registerParticipant(...);  
void displayParticipants(...);  
void genderWiseCount(...);  
void eventWiseCount(...);  
void maximumEventRegistrant(...);
```

You may create additional functions if required.

Programming Constraints

- Use nested structures.
- Use an array of structures.
- Use malloc()/realloc() for dynamic memory allocation.
- Use free() before termination.
- Do not use a fixed-size array for the total number of participants.
- Do not maintain a separate event-registration count array.
- Calculate event-wise statistics by traversing the participant records.
- Validate all registration rules.
- Produce clear and properly formatted output.

Minimum Test Cases

Demonstrate the following during execution:

1. Successful participant registration.
2. Attempt to register for more than 3 events.
3. Duplicate Registration ID.
4. Duplicate event registration.
5. Invalid event code.
6. Invalid gender.
7. Attempt to register for a full event when 6 participants are already registered.
8. Successful registration in another event when one event is full.
9. Gender-wise statistics.
10. Event-wise statistics.
11. Participant(s) with maximum event registrations.
12. Proper memory release during exit.

Expected Outcome:

The program should provide a reliable TechTatva Fest registration system that dynamically manages participants, enforces both **participant-level and event-level capacity constraints**, prevents duplicate registrations, and generates statistics directly from the dynamically allocated array of structures.

Evaluation Pattern:

Question No	Writeup (2 Marks)	Execution (2 Marks)	Total Marks=20 (W+E)
Case-1			
Case-2			
Case-3			
Case-4			
Case-5			

*****THE END*****

COMPLETE SAMPLE OUTPUT

TECHTATVA FEST REGISTRATION SYSTEM

TECHTATVA FEST REGISTRATION SYSTEM

1. Register Participant
2. Display Participants
3. Gender-wise Statistics
4. Event-wise Statistics
5. Maximum Event Registrant
6. Exit

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 101

Enter Name: RAHUL

Enter Gender (M/F/O): M

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 3

Enter event code: 1

SAAHITI registered successfully.

Enter event code: 2

IKYA registered successfully.

Enter event code: 3

ROBOWARS registered successfully.

Participant registered successfully.

TECHTATVA FEST REGISTRATION SYSTEM

1. Register Participant
2. Display Participants
3. Gender-wise Statistics
4. Event-wise Statistics
5. Maximum Event Registrant
6. Exit

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 102

Enter Name: SAHANA

Enter Gender (M/F/O): F

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 3

Enter event code: 4

CODE MARATHON registered successfully.

Enter event code: 5

AIrena registered successfully.

Enter event code: 6

Quantum Leap registered successfully.

Participant registered successfully.

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 103

Enter Name: Aarush

Enter Gender (M/F/O): m

Invalid gender.

Please enter M, F or O.

Enter Gender (M/F/O): M

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 3

Enter event code: 1

SAAHITI registered successfully.

Enter event code: 2

IKYA registered successfully.

Enter event code: 4

CODE MARATHON registered successfully.

Participant registered successfully.

```
=====
TECHTATVA FEST
REGISTRATION SYSTEM
=====
```

1. Register Participant
2. Display Participants
3. Gender-wise Statistics
4. Event-wise Statistics
5. Maximum Event Registrant
6. Exit

```
=====
```

Enter your choice: 1

```
===== PARTICIPANT REGISTRATION =====
```

Enter Registration ID: 104

Enter Name: Mahisha

Enter Gender (M/F/O): F

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 3

Enter event code: 1

SAAHITI registered successfully.

Enter event code: 2

IKYA registered successfully.

Enter event code: 6

Quantum Leap registered successfully.

Participant registered successfully.

```
=====
TECHTATVA FEST
REGISTRATION SYSTEM
=====
```

1. Register Participant
2. Display Participants
3. Gender-wise Statistics
4. Event-wise Statistics
5. Maximum Event Registrant
6. Exit

```
=====
```

Enter your choice: 1

```
===== PARTICIPANT REGISTRATION =====
```

Enter Registration ID: 105

Enter Name: AADITI

Enter Gender (M/F/O): F

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 3

Enter event code: 1

SAAHITI registered successfully.

Enter event code: 4

CODE MARATHON registered successfully.

Enter event code: 5

AIrena registered successfully.

Participant registered successfully.

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 106

Enter Name: INDRANEEL

Enter Gender (M/F/O): M

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 3

Enter event code: 3

ROBOWARS registered successfully.

Enter event code: 4

CODE MARATHON registered successfully.

Enter event code: 6

Quantum Leap registered successfully.

Participant registered successfully.

TECHTATVA FEST
REGISTRATION SYSTEM

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
-
-

Enter your choice: 2

PARTICIPANT DETAILS

Registration ID : 101
Name : RAHUL
Gender : M
Events Registered: 3
Registered Events:
1. SAAHITI
2. IKYA
3. ROBOWARS

Registration ID : 102
Name : SAHANA
Gender : F
Events Registered: 3
Registered Events:
1. CODE MARATHON
2. AIrena
3. Quantum Leap

Registration ID : 103
Name : Aarush
Gender : M
Events Registered: 3
Registered Events:
1. SAAHITI
2. IKYA
3. CODE MARATHON

Registration ID : 104
Name : Mahisha
Gender : F
Events Registered: 3
Registered Events:
1. SAAHITI
2. IKYA
3. Quantum Leap

Registration ID : 105
Name : AADITI
Gender : F

Events Registered: 3
Registered Events:
1. SAAHITI
2. CODE MARATHON
3. AIrena

Registration ID : 106
Name : INDRANEEL
Gender : M
Events Registered: 3
Registered Events:
1. ROBOWARS
2. CODE MARATHON
3. Quantum Leap

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 3

=====

GENDER-WISE STATISTICS

=====

Male Participants : 3
Female Participants : 3
Other Participants : 0
Total Participants : 6

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 4

=====

EVENT-WISE STATISTICS

=====

Event	Registered	Capacity	Available	Status
SAAHITI	4	6	2	AVAILABLE
IKYA	3	6	3	AVAILABLE
ROBOWARS	2	6	4	AVAILABLE
CODE MARATHON	4	6	2	AVAILABLE
AIrena	2	6	4	AVAILABLE

Quantum Leap	3	6	3	AVAILABLE
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Total Event Registrations: 18

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TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 5

=====

MAXIMUM EVENT REGISTRATION

=====

Maximum Number of Events : 3

Reg ID	Name	Events Registered
101	RAHUL	3
102	SAHANA	3
103	Aarush	3
104	Mahisha	3
105	AADITI	3
106	INDRANEEL	3

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 101

Registration ID already exists.
Duplicate Registration ID is not allowed.

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
2. Display Participants
3. Gender-wise Statistics
4. Event-wise Statistics
5. Maximum Event Registrant

6. Exit

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 107

Enter Name: PRIYANKA

Enter Gender (M/F/O): F

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 4

Maximum 3 events are allowed.

Enter number of events (1-3): 3

Enter event code: 1

SAAHITI registered successfully.

Enter event code: 4

CODE MARATHON registered successfully.

Enter event code: 5

AIrena registered successfully.

Participant registered successfully.

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 108

Enter Name: VARUN

Enter Gender (M/F/O): M

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 1

Enter event code: 8

Invalid event code.

Please enter a value between 1 and 6.

Enter event code: 2

IKYA registered successfully.

Participant registered successfully.

TECHTATVA FEST
REGISTRATION SYSTEM

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
-
-

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 109

Enter Name: KAVYA

Enter Gender (M/F/O): F

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 1

Enter event code: 1

SAAHITI registered successfully.

Participant registered successfully.

TECHTATVA FEST
REGISTRATION SYSTEM

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
-
-

Enter your choice: 1

===== PARTICIPANT REGISTRATION =====

Enter Registration ID: 110

Enter Name: ROHIT
Enter Gender (M/F/O): M

Available Events:

1. SAAHITI
2. IKYA
3. ROBOWARS
4. CODE MARATHON
5. AIrena
6. Quantum Leap

Enter number of events (1-3): 1

Enter event code: 1

SAAHITI is FULL.
Maximum capacity of 6 participants has been reached.
Registration for this event is not allowed.

Enter event code: 5
AIrena registered successfully.

Participant registered successfully.

TECHTATVA FEST REGISTRATION SYSTEM

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
-
-

Enter your choice: 2

PARTICIPANT DETAILS

Registration ID : 101
Name : RAHUL
Gender : M
Events Registered: 3
Registered Events:
1. SAAHITI
2. IKYA
3. ROBOWARS

Registration ID : 102
Name : SAHANA
Gender : F
Events Registered: 3
Registered Events:
1. CODE MARATHON
2. AIrena
3. Quantum Leap

Registration ID : 103
Name : Aarush
Gender : M
Events Registered: 3
Registered Events:
1. SAAHITI
2. IKYA
3. CODE MARATHON

Registration ID : 104
Name : Mahisha
Gender : F
Events Registered: 3
Registered Events:
1. SAAHITI
2. IKYA
3. Quantum Leap

Registration ID : 105
Name : AADITI
Gender : F
Events Registered: 3
Registered Events:
1. SAAHITI
2. CODE MARATHON
3. AIrena

Registration ID : 106
Name : INDRANEEL
Gender : M
Events Registered: 3
Registered Events:
1. ROBOWARS
2. CODE MARATHON
3. Quantum Leap

Registration ID : 107
Name : PRIYANKA
Gender : F
Events Registered: 3
Registered Events:
1. SAAHITI
2. CODE MARATHON
3. AIrena

Registration ID : 108
Name : VARUN
Gender : M
Events Registered: 1
Registered Events:
1. IKYA

Registration ID : 109
Name : KAVYA

Gender : F
Events Registered: 1
Registered Events:
1. SAAHITI

Registration ID : 110
Name : ROHIT
Gender : M
Events Registered: 1
Registered Events:
1. AIrena

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 3

=====

GENDER-WISE STATISTICS

=====

Male Participants : 5
Female Participants : 5
Other Participants : 0
Total Participants : 10

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 4

=====

EVENT-WISE STATISTICS

=====

Event	Registered	Capacity	Available	Status
SAAHITI	6	6	0	FULL
IKYA	4	6	2	AVAILABLE
ROBOWARS	2	6	4	AVAILABLE
CODE MARATHON	5	6	1	AVAILABLE
AIrena	4	6	2	AVAILABLE

Quantum Leap	3	6	3	AVAILABLE
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Total Event Registrations: 24

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TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 5

=====

MAXIMUM EVENT REGISTRATION

=====

Maximum Number of Events : 3

Reg ID	Name	Events Registered
101	RAHUL	3
102	SAHANA	3
103	Aarush	3
104	Mahisha	3
105	AADITI	3
106	INDRANEEL	3
107	PRIYANKA	3

=====

TECHTATVA FEST
REGISTRATION SYSTEM

=====

1. Register Participant
 2. Display Participants
 3. Gender-wise Statistics
 4. Event-wise Statistics
 5. Maximum Event Registrant
 6. Exit
- =====

Enter your choice: 6

Dynamically allocated memory released.
Thank you for using TechTatva Fest System.

Recommended final execution sequence

- 1 → Register 101 RAHUL
- 1 → Register 102 SAHANA
- 1 → Register 103 Aarush
- 1 → Register 104 Mahisha
- 1 → Register 105 AADITI
- 1 → Register 106 INDRANEEL

- 2 → Display Participants
- 3 → Gender-wise Statistics
- 4 → Event-wise Statistics
- 5 → Maximum Event Registrant

- 1 → Attempt duplicate Registration ID 101

- 1 → Attempt more than 3 events
 - Enter 4
 - Display "Maximum 3 events are allowed."
 - Re-enter valid 3 events

- 1 → Attempt invalid gender
 - Enter invalid gender
 - Display "Invalid gender."
 - Re-enter valid gender

- 1 → Attempt invalid event code
 - Enter 8
 - Display "Invalid event code."
 - Enter valid event code

- 1 → Register additional participants
 - Register 109 KAVYA
 - Select SAAHITI
 - SAAHITI becomes FULL (6/6)

- 1 → Attempt 7th registration for SAAHITI
 - Display FULL message
 - Select another available event
 - Successful registration

- 4 → Display updated event statistics

- 5 → Display updated maximum event registrant

- 6 → Release memory and exit